

COL774: MACHINE LEARNING

Assignment 4: Learning to Solve Mazes

Shivaani Hari 2022MT11273
Shrenik Sakala 2022MT11920

Introduction

Our goal is to use sequence-to-sequence (Seq2Seq) models for tasks involving spatial reasoning, specifically for solving mazes. Given a tokenized representation of the maze structure (adjacency list) along with a start and end point, the model must predict the optimal sequence of coordinates forming the solution path.

We explore two distinct architectures for this task: a Recurrent Neural Network (RNN) with Bahdanau Attention and a Transformer-based model. The models are trained in an end-to-end fashion, learning to map the variable-length input sequence (maze description) to the output sequence (solution path). Below are the key implementation details and hyperparameters used for training:

- **Optimizer:** Adam
- **Loss Function:** Cross-Entropy Loss
- **Learning Rate:** 1×10^{-4}
- **Batch Size:** 32
- **Number of Epochs:** 20

By utilizing these architectures, we aim to analyze the capability of sequential models to learn the complex structural dependencies and spatial planning required for pathfinding.

Recurrent Neural Networks (RNNs)

Values of training and validation losses, token and sequence accuracies were obtained and recorded across different epochs.

Epoch	Train Loss	Val Loss	Train Tok Acc	Val Tok Acc	Train Seq Acc	Val Seq Acc
1	2.0614	2.3054	0.391	0.350	0.029	0.147
2	1.5653	2.1801	0.521	0.386	0.149	0.204
3	1.4816	2.0563	0.558	0.430	0.195	0.259
4	1.2953	1.7221	0.635	0.550	0.265	0.387
5	1.0335	1.5451	0.728	0.618	0.416	0.519
6	0.8801	1.5002	0.771	0.635	0.504	0.541
7	0.7885	1.3851	0.793	0.648	0.539	0.553
8	0.7456	1.3252	0.801	0.661	0.559	0.562
9	0.6895	1.2570	0.812	0.671	0.569	0.571
10	0.6458	1.2108	0.824	0.682	0.585	0.575
11	0.6151	1.2003	0.830	0.678	0.595	0.562
12	0.5823	1.1706	0.837	0.687	0.604	0.570
13	0.5476	1.1202	0.847	0.705	0.615	0.589
14	0.5250	1.0719	0.851	0.709	0.621	0.592
15	0.5048	1.0186	0.855	0.714	0.629	0.580
16	0.4828	1.0423	0.861	0.709	0.635	0.580
17	0.4638	0.9885	0.865	0.725	0.641	0.592
18	0.4348	0.9832	0.873	0.731	0.649	0.600
19	0.4186	1.0623	0.878	0.729	0.658	0.594
20	0.4065	1.0159	0.880	0.733	0.660	0.590

Table 1: Train and Validation Metrics (RNNs)

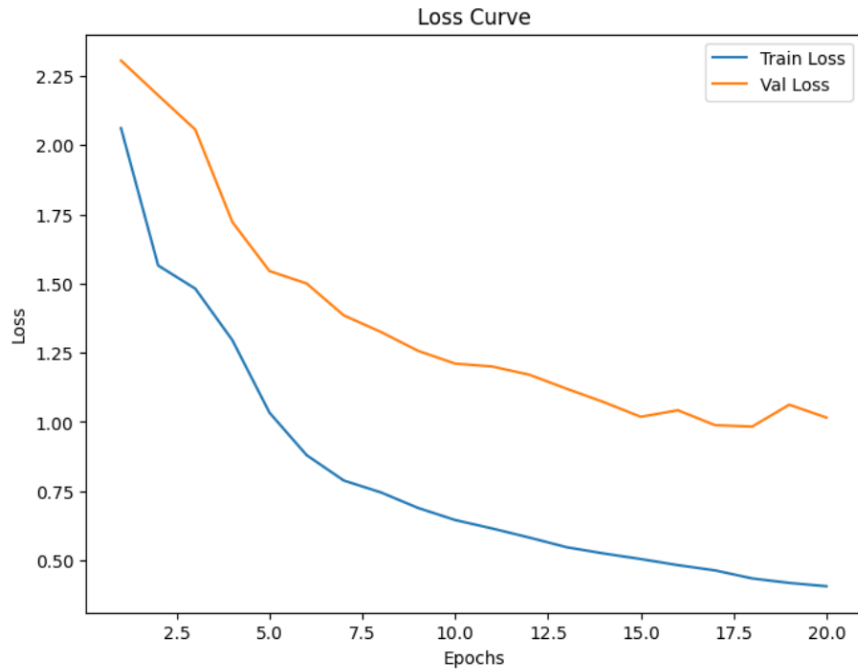


Figure 1: Training and Validation Losses vs Number of Epochs

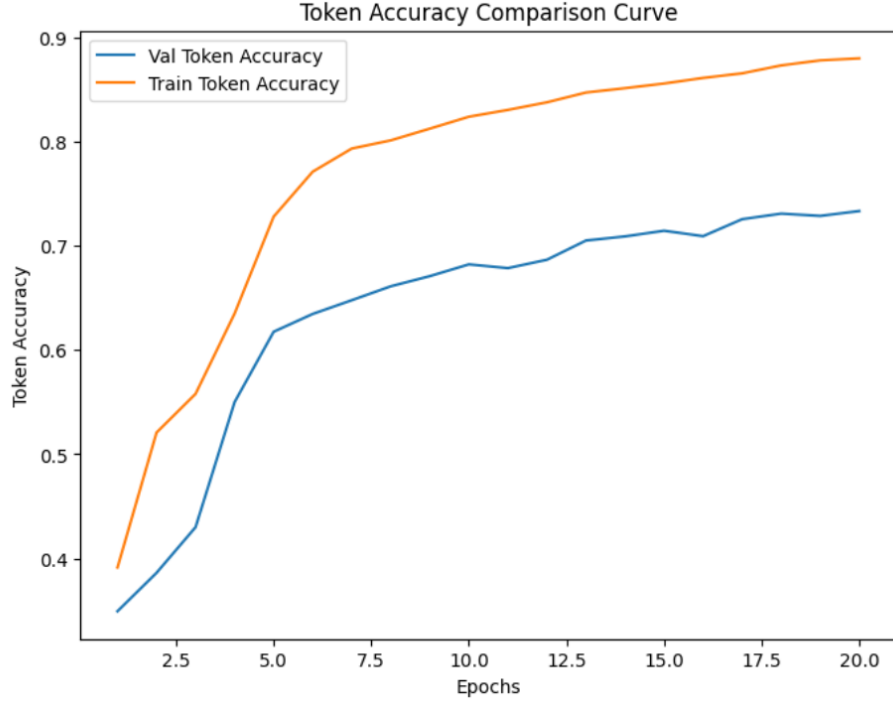


Figure 2: Training and Validation Token Accuracies vs Number of Epochs

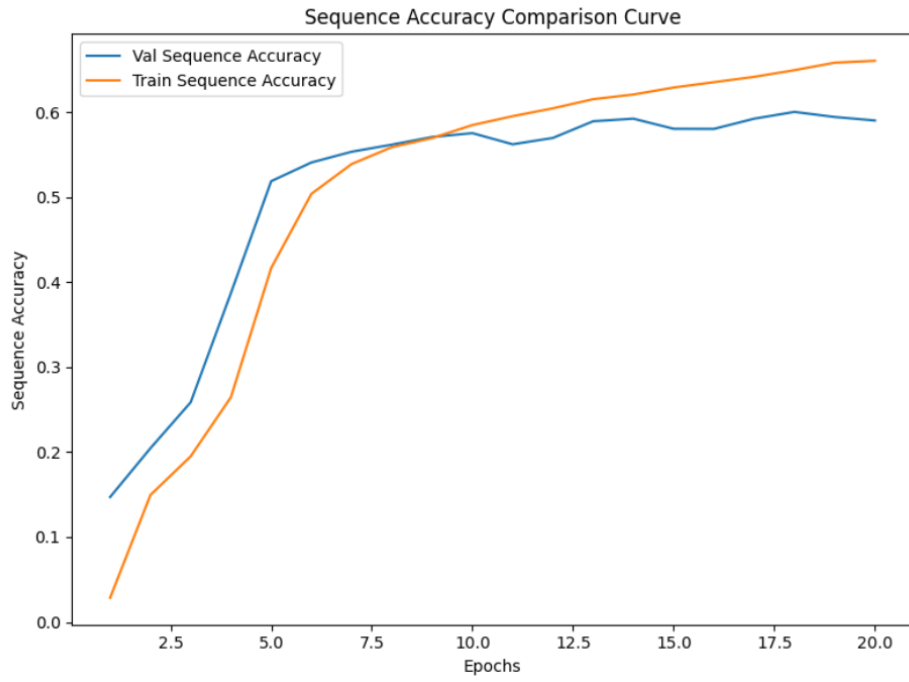


Figure 3: Training and Validation Sequence Accuracies vs Number of Epochs

We also generate the paths for 5 random mazes from the validation dataset.

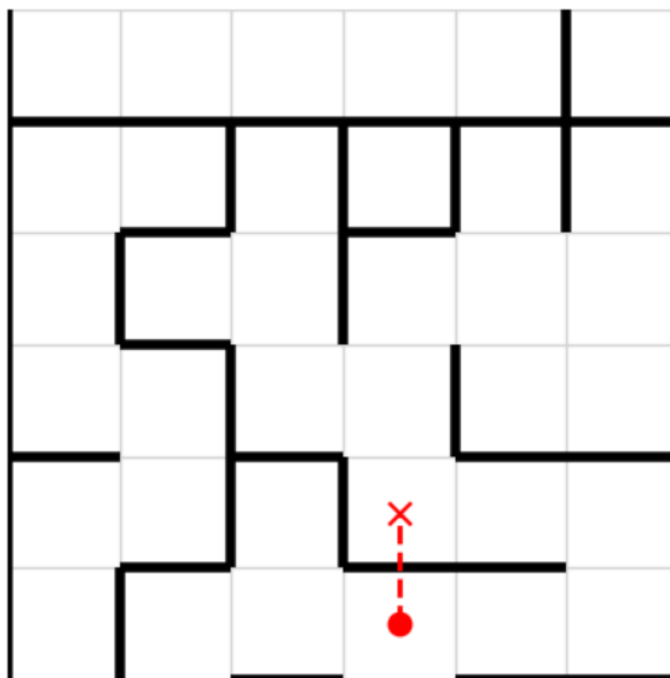


Figure 4: Random Sample 1- Path by RNN

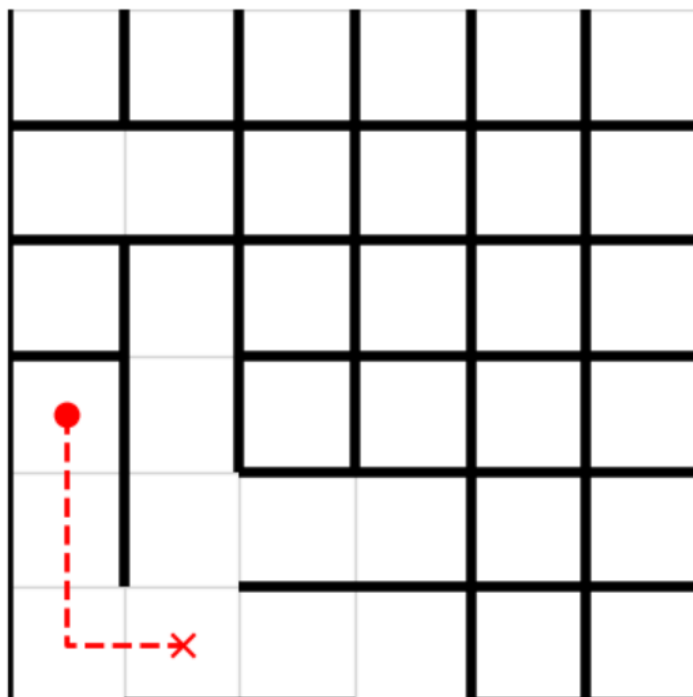


Figure 5: Random Sample 2- Path by RNN

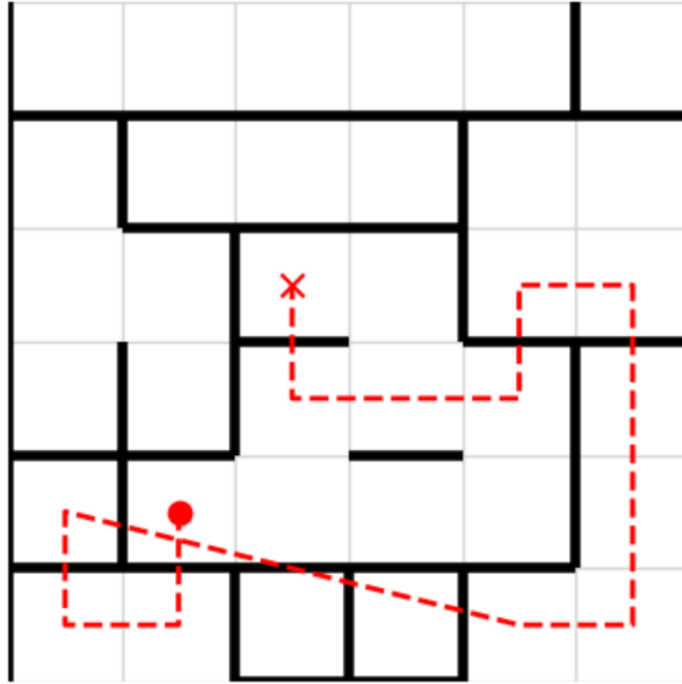


Figure 6: Random Sample 3- Path by RNN

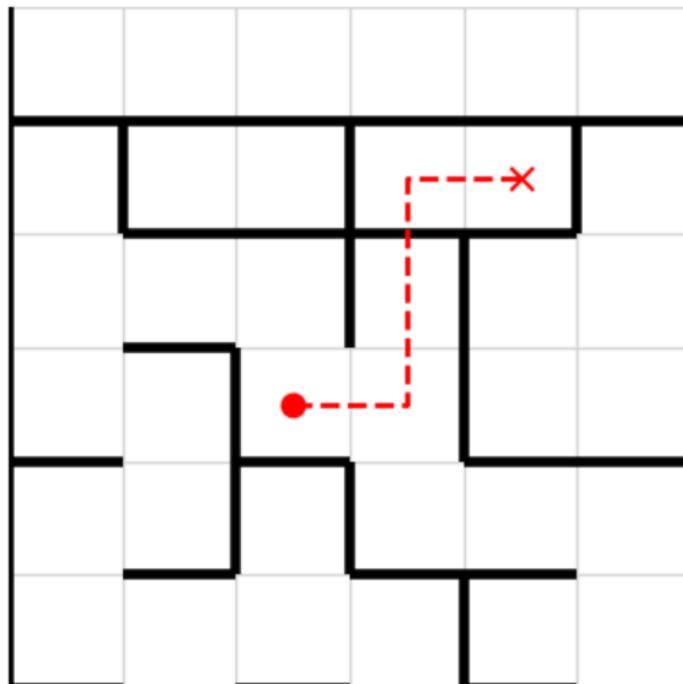


Figure 7: Random Sample 4- Path by RNN

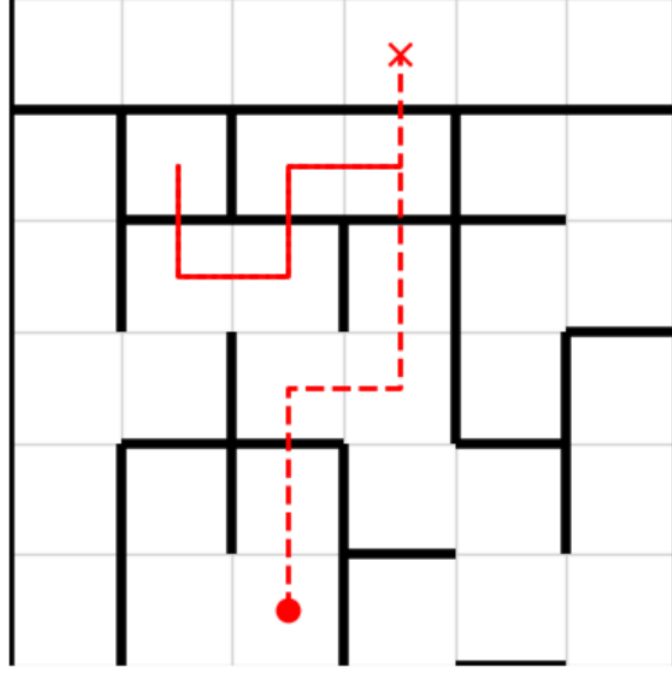


Figure 8: Random Sample 5- Path by RNN

Then, we evaluated the model on both train and test datasets, to calculate the final F1 scores on each of them.

- Train F1 score= 0.8602
- Test F1 score= 0.7554

The corresponding sequence accuracies that were obtained were-

- Train= 74.17%
- Test= 62.41%

Transformer Architecture

Values of training and validation losses, token and sequence accuracies were obtained and recorded across different epochs.

Epoch	Train Loss	Val Loss	Train Tok Acc	Val Tok Acc	Train Seq Acc	Val Seq Acc
1	1.3197	0.8476	0.514	0.654	0.002	0.003
2	0.8702	0.6827	0.645	0.709	0.009	0.017
3	0.7364	0.5669	0.693	0.753	0.028	0.083
4	0.6308	0.4909	0.733	0.782	0.067	0.130
5	0.5549	0.4376	0.763	0.806	0.098	0.153
6	0.5073	0.4141	0.782	0.815	0.117	0.167
7	0.4749	0.4089	0.795	0.818	0.130	0.176
8	0.4486	0.3661	0.806	0.835	0.145	0.191
9	0.4284	0.3610	0.814	0.836	0.154	0.198
10	0.4104	0.354	0.822	0.840	0.164	0.204
11	0.3952	0.3385	0.827	0.845	0.172	0.220
12	0.3806	0.3304	0.834	0.851	0.186	0.234
13	0.3654	0.3074	0.840	0.861	0.202	0.270
14	0.3442	0.2860	0.849	0.871	0.228	0.305
15	0.3252	0.2663	0.857	0.878	0.258	0.342
16	0.3100	0.2710	0.863	0.876	0.279	0.342
17	0.2967	0.2610	0.868	0.880	0.296	0.354
18	0.2856	0.2514	0.872	0.884	0.309	0.368
19	0.2767	0.2513	0.876	0.883	0.324	0.363
20	0.2686	0.2426	0.879	0.886	0.324	0.377

Table 2: Train and Validation Metrics (Transformers)

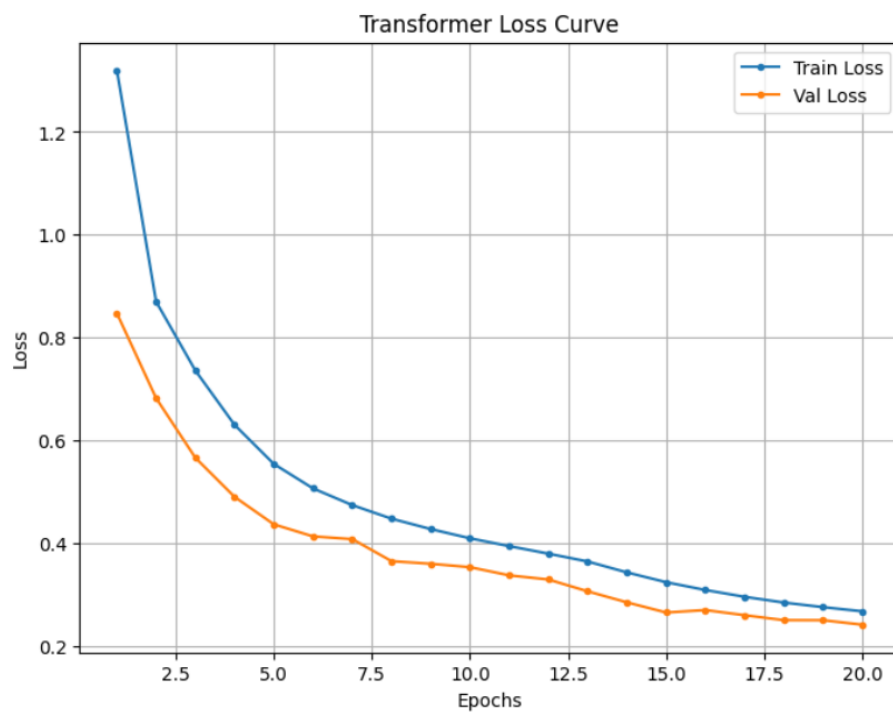


Figure 9: Training and Validation Loss vs Number of Epochs

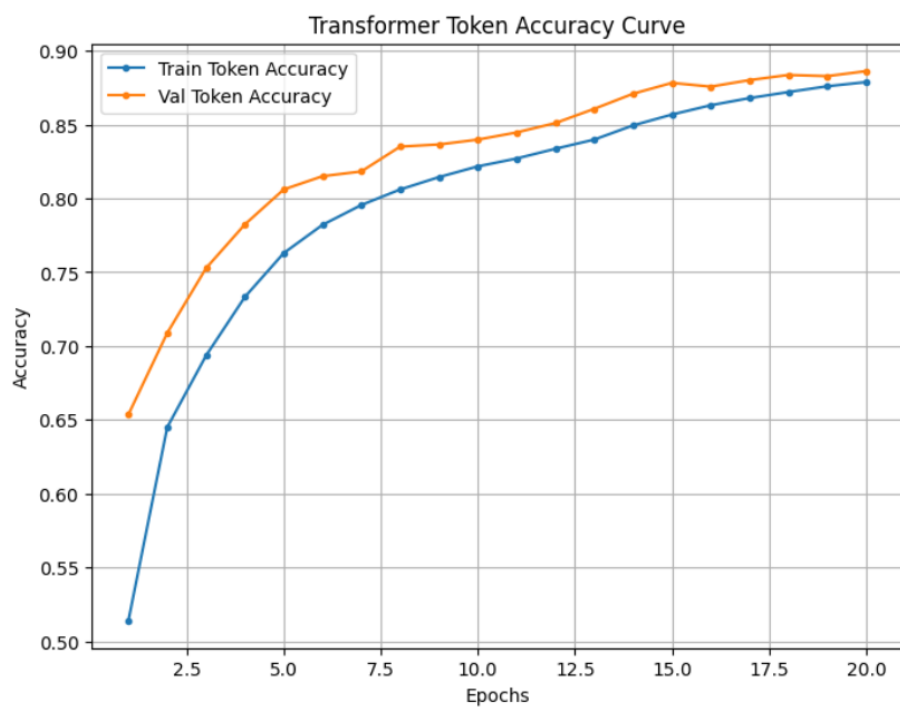


Figure 10: Training and Validation Token Accuracies vs Number of Epochs

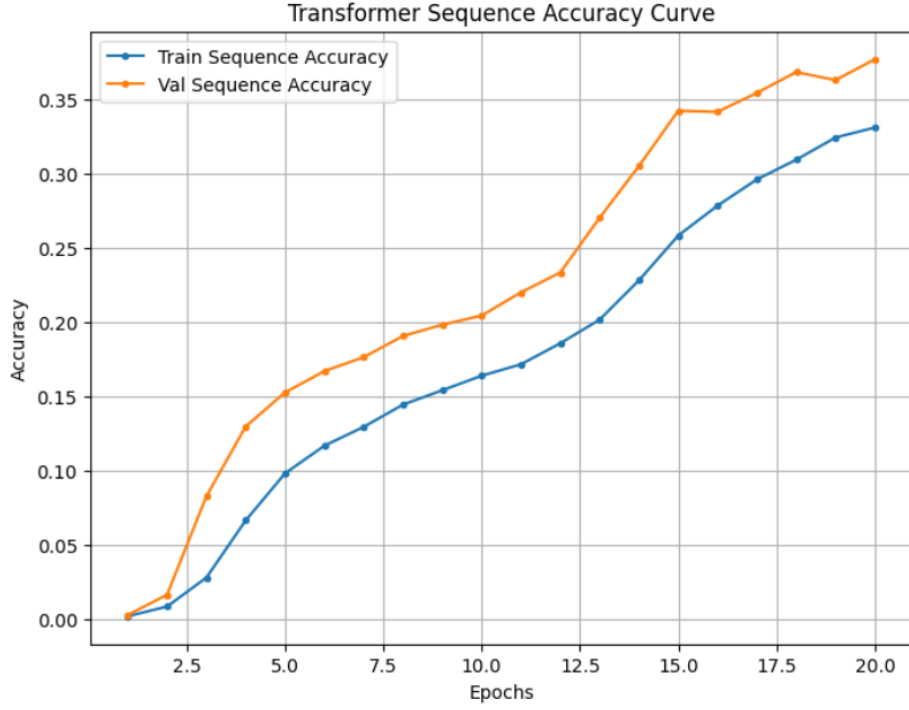


Figure 11: Training and Validation Sequence Accuracies vs Number of Epochs

We also generate the paths for 5 random mazes from the validation dataset.

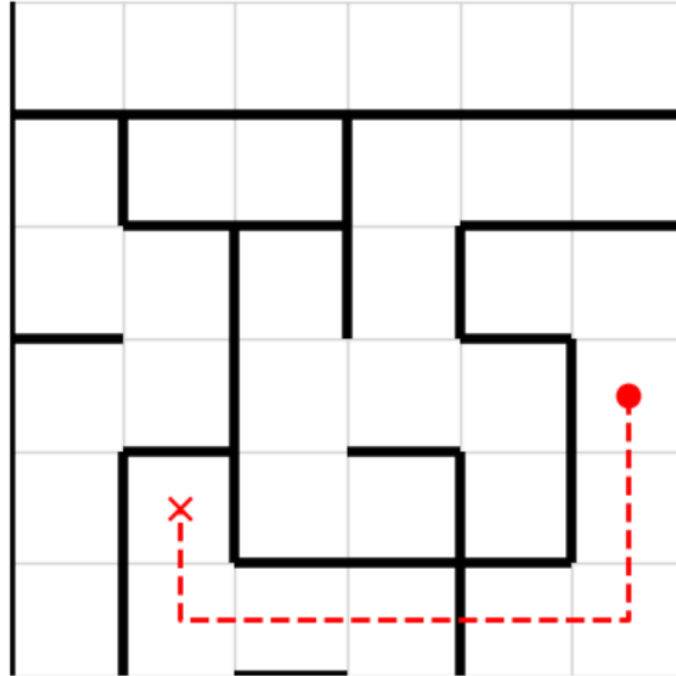


Figure 12: Random Sample 1- Path by Transformer

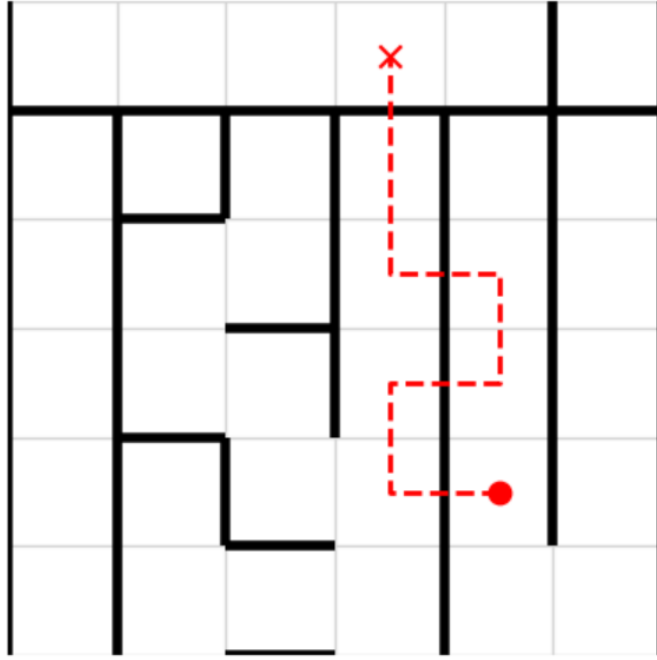


Figure 13: Random Sample 2- Path by Transformer

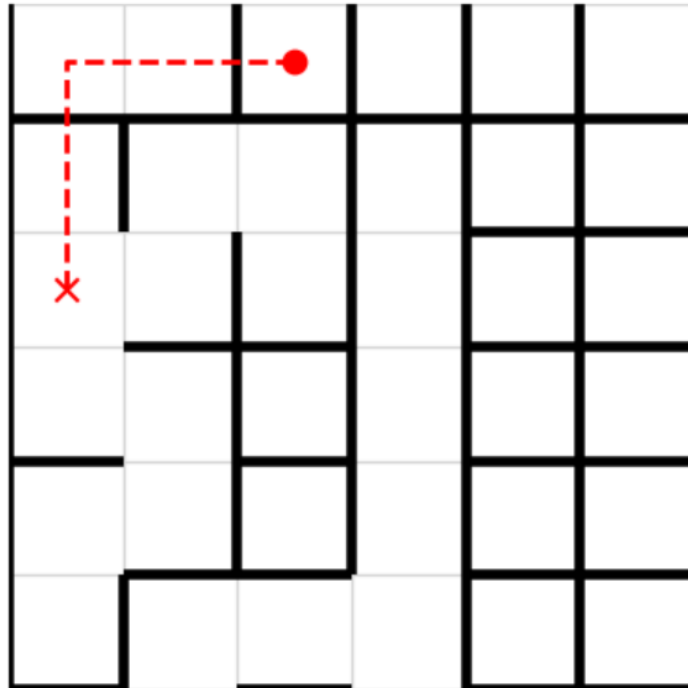


Figure 14: Random Sample 3- Path by Transformer

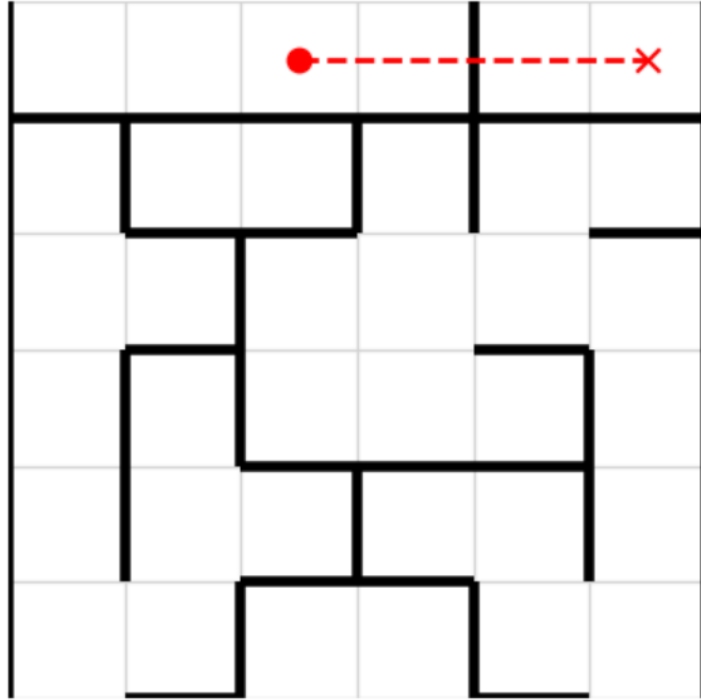


Figure 15: Random Sample 4- Path by Transformer

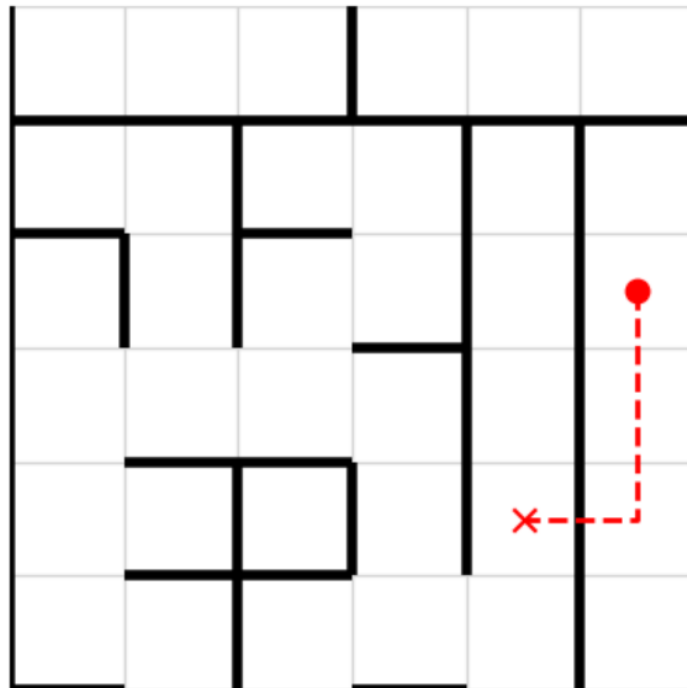


Figure 16: Random Sample 5- Path by Transformer

Then, we evaluated the model on both train and test datasets, to calculate the final F1 scores (token) on each of them.

- Train F1 score= 0.4784

- Test F1 score= 0.4491

The corresponding sequence accuracies that were obtained were-

- Train= 40.27%
- Test= 36.18%